



Indira College of Commerce and Science, Pune
Department of Computer Science
T.Y.B.Sc.(Computer Science) Project

Synopsis

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Project Title: Employee Productivity & Workload Balancing Platform

Project Description in Brief (Problem Definition)

In many organizations, task allocation is often done without analyzing employee workload, capacity, or deadline pressure. This leads to uneven work distribution, employee burnout, missed deadlines, and reduced productivity. Managers lack a centralized system to monitor workload, track task progress, and identify performance risks in real time.

The proposed **Employee Productivity & Workload Balancing Platform** is a web-based application designed to track employee tasks, workloads, and deadlines in a structured and data-driven manner. The system enables efficient task assignment, monitors employee workload utilization, and identifies deadline risks and productivity patterns.

This platform improves decision-making for managers, enhances transparency, ensures balanced workload distribution, and increases overall team productivity while reducing operational inefficiencies.

Technology Used (Software and Hardware Requirements)

Software Requirements:

1. Java (Spring Boot)
2. PostgreSQL Database
3. RESTful APIs
4. Postman (API Testing & Documentation)
5. HTML, CSS (Frontend)
6. Visual Studio Code / IntelliJ IDEA

Hardware Requirements:

1. Computer / Laptop
2. Minimum 8 GB RAM
3. Windows / Linux Operating System

Module Details

1. Employee & Task Management Module

This module handles employee details, task creation, and workload tracking.

Sub-modules:

- Employee Registration and Profile Management
- Task Creation and Assignment by Manager
- Task Status Tracking (Not Started / In Progress / Completed)
- Workload Calculation based on assigned tasks and capacity
- Deadline Monitoring and Risk Detection

This module ensures proper task tracking and balanced workload distribution.

2. Productivity & Analytics Module

This module analyzes employee performance and generates productivity insights.

Sub-modules:

- Workload Analysis (Overutilized / Underutilized Employees)
- Deadline Risk Indicators
- Productivity Score Calculation
- Task Completion and Performance Reports
- Workload Distribution Monitoring

This module helps managers make data-driven decisions and improve team efficiency.

3. API & System Management Module

This module handles backend operations, API management, and system reliability.

Sub-modules:

- RESTful API Development for task and workload management
- API Testing and Documentation using Postman
- Backend Deployment and Configuration
- Secure Data Storage using PostgreSQL

- System Monitoring and Maintenance

This module ensures smooth system operation and reliable data handling.

Functional Details

1. User Authentication and Role Management

- Provides secure login and authentication for users
- Assigns roles such as Manager or Employee based on credentials
- Grants system access according to assigned role

2. Employee Management

- Manages employee profile information and role details
- Stores employee workload capacity
- Maintains structured employee records
- Supports accurate workload calculations

3. Task Creation and Assignment

- Allows managers to create tasks with title and description
- Sets task priority, estimated effort, and deadline
- Assigns tasks to employees for execution and tracking

4. Task Status Tracking

- Enables employees to update task status
- Supports status values: Not Started, In Progress, Completed
- Allows real-time monitoring of work progress and completion

5. Workload Calculation and Monitoring

- Calculates total assigned workload for each employee

- Compares assigned workload with working capacity
- Identifies balanced, underutilized, or overloaded conditions

6. Deadline Risk Detection

- Analyzes task deadlines and workload status
- Identifies potential delivery risks
- Categorizes tasks as Low, Medium, or High Risk

7. Productivity Analysis and Reporting

- Generates productivity metrics based on task completion rate
- Tracks deadline adherence and workload handling
- Provides productivity reports and performance insights to managers

8. API Management and System Operations

- Uses RESTful APIs for managing tasks and workload data
- Handles productivity metrics through backend APIs
- Tests and documents APIs using Postman for reliability and correctness

9. Data Storage and Database Management

- Stores employee, task, workload, and performance data securely
- Uses PostgreSQL for structured data storage
- Enables efficient data retrieval and management

Signature of Student

Receivers Signature

Date:

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